

Higher Criticism for Adaptive Retrieval in Retrieval-Augmented Generation: A Controlled-Environment Study

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Final Project Report

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Abstract

Retrieval-Augmented Generation (RAG) pipelines almost always retrieve a *fixed* number of passages per query, wasting context on easy questions and starving hard ones. We study *Higher Criticism* (HC), a statistical test for rare and weak signals, as a query-adaptive cutoff that decides *how many* documents to keep. We first integrated HC into three established pipelines (Adaptive-K, Adaptive-RAG, FlashRAG): HC was competitive-to-best where the number of relevant documents is large and variable, but every pipeline and public benchmark carried a confound — a keyword-only backend, a tiny clustered corpus that invalidated the null distribution, or a fixed candidate pool — that prevented a clean test of HC. We therefore built a family of *controlled* Amazon retrieval datasets with fully known ground truth and a single tunable difficulty knob. On the clean **Category-Key** corpus HC beats every fixed top- k on F1 (0.893 vs. 0.783) by tracking each query’s true relevance count. On harder **compound** (category + price) queries HC breaks — 55/135 queries return $k = 0$ — and the controlled design lets us attribute each failure to one of three mechanisms: embedding limitation, *calibration mismatch*, and signal dilution. Calibration mismatch (a per-query noise scale narrower than the global null, which inflates p -values) is the dominant, fixable cause, and is exactly the “ p -values too high” problem raised in our presentation feedback. Our main contribution, **rescale.trim**, is a label-free per-query recalibration applied before a single global null; it recovers 30 of 31 fixable queries and makes HC the best of eight selection rules (best F1 0.209), generalizing to a 100K-document electronics corpus and through the full RAG chain.

1 Background and Motivation

Retrieval-Augmented Generation (RAG) (Lewis et al. 2020) augments a Large Language Model (LLM) with passages pulled from an external corpus, usually via dense similarity search over an approximate-nearest-neighbour index. The dominant interface is *Top-K*: fix an integer k and pass the k most similar passages to the generator. A single k cannot simultaneously serve a fact-lookup that needs one passage, a multi-hop question that needs three, and a database-style

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[Figure placeholder: “Database retrieval as sparse signal detection.” Sorted similarity scores for one query; a few relevant documents (signal) sit above a bulk of irrelevant documents (null). The HC cutoff marks where signal gives way to noise.]

Figure 1: Retrieval as sparse signal detection: HC locates the boundary between a few relevant documents and the irrelevant bulk.

aggregation that needs dozens. Too small a k discards evidence; too large a k inflates context length, latency, and cost, and degrades generation through distractor noise.

1.1 Database Retrieval as Sparse Signal Detection

The starting observation of this project is that *choosing k is a detection problem*. For a given query, the corpus contains a small number of *relevant* documents hidden among a large majority of irrelevant ones. After embedding and similarity search, relevant documents tend to score higher than irrelevant ones, but only by a margin that varies from query to query. Reading the sorted similarity scores as a sample, the question “how many of the top candidates are genuinely relevant?” is precisely the classical question of detecting a *sparse* set of signals against a noisy null — the regime that Higher Criticism was designed for. Figure 1 illustrates this view: the cutoff k is the boundary where the score distribution stops looking anomalous and settles into noise.

1.2 What a p -value Means Here, and How HC Selects Documents

To turn similarities into a detection test we need a *null distribution*: the distribution of similarity scores for *irrelevant* (query, document) pairs. Given that null, each candidate’s score s_i is mapped to a z -score and then to a p -value p_i — the probability that an irrelevant document would score at least as high. A genuinely relevant document is anomalously similar, so it receives a *tiny* p -value; an irrelevant document receives a p -value that is roughly uniform on $[0, 1]$. HC then scans the sorted p -values and selects the cutoff at which the observed p -values deviate *most* from the uniform-noise expectation (Section 2.1). The number of documents kept, k^* ,

is chosen per query. Two facts drive the rest of this report: the test is only as good as the null, and the quantiles HC compares against are the quantiles of *that null* — so a miscalibrated null silently corrupts every p -value.

1.3 Existing Adaptive Approaches

Two families of remedies exist. *Adaptive-K* (Labs 2025) chooses k by locating the largest gap in the sorted similarity scores — cheap, but brittle when the score distribution is flat. *Adaptive-RAG* (Jeong et al. 2024) trains a classifier that routes each query to {no retrieval, single-step retrieval, iterative IRCot} — accurate on multi-hop questions but paying many retriever and generator calls. Neither asks the statistical question “where does relevance end?” directly; HC does.

1.4 Research Question

On what kind of data, and under what calibration, can an HC-driven cutoff beat fixed- k and other adaptive selection rules — and when and why does it fail? Answering this cleanly requires a setting we control. The body of the report builds that setting and uses it to both validate HC and diagnose, then repair, its dominant failure mode.

2 Methodology

2.1 The HC Retrieval Algorithm

For each query q the retriever returns the top N candidates with similarities $s_1 \geq \dots \geq s_N$. Each s_i is standardized under the null to a z -score and converted to a p -value p_i . Sorting $p_{(1)} \leq \dots \leq p_{(N)}$, the HC statistic at position i is

$$HC_i = \sqrt{N} \frac{i/N - p_{(i)}}{\sqrt{p_{(i)}(1 - p_{(i)})}},$$

which measures, in standard-error units, how far the i -th smallest p -value falls below its uniform-null expectation i/N . The adaptive cutoff is $k^* = \arg \max_{i \leq \gamma N} HC_i$, with γ a small upper-fraction hyperparameter. If $\max_i HC_i \leq 0$ the test declares “no signal” and returns $k^* = 0$. The comparison value i/N is the uniform quantile *under the null*: the entire procedure presumes that irrelevant-document p -values are uniform, which we return to in Section 4.5.

2.2 Pipelines and Selection Rules Compared

We treat HC as a drop-in *cutoff stage* and compare it against the selection rules in Table 1, spanning learned routers, gap heuristics, classical multiple-testing corrections, and clustering. Statistical-test rules (HC, Berk–Jones, Bonferroni, BH) consume identical p -values; Adaptive-K and CAR read similarities directly.

2.3 System Architecture

Figure 2 shows one full RAG pipeline with HC as the cutoff stage. *Offline*: chunk, embed, index, and calibrate a null distribution. *Online*: embed the query, retrieve a candidate pool, apply the HC cutoff, build the prompt, generate, and evaluate both retrieval and generation.

[Figure placeholder: HC-RAG pipeline. Offline: chunk $\rightarrow$ embed $\rightarrow$ FAISS index $\rightarrow$ null calibration. Online: query $\rightarrow$ ANN pool $\rightarrow$ HC cutoff (k^*) $\rightarrow$ prompt $\rightarrow$ LLM $\rightarrow$ IR + generation eval.]

Figure 2: One RAG pipeline end to end, with HC as a drop-in cutoff.

2.4 Datasets

The **controlled Amazon** datasets are the core of this study, all derived from McAuley–Lab/Amazon–Reviews–2023 with OpenAI text-embedding-3-small (1536d) in a FAISS IndexFlatIP: **Category-Key** (44 “largest cities in X ” queries over 726 passages, variable true K , BGE embeddings); **Amazon-Compound** (80K documents across three departments, 135 category+price queries, 4.8% relevant); and **Electronics-100K** (100K single-department documents, 44 queries, 1.2% relevant, 44.5% uncapped hard negatives). Each has fully known relevance tiers and a per-query true K .

For the earlier exploration (Section 3) we used the public benchmarks **CrossEntityQA** (our sparse multi-entity benchmark; 13K docs, 2–20 relevant per query), **MuSiQue** (Trivedi et al. 2022), **SQuAD**, **HotpotQA** (Yang et al. 2018), **HoloBench** (Maekawa and others 2024), and **BEIR SciFact/FiQA** (Thakur et al. 2021; Wadden et al. 2020).

2.5 Setup and Metrics

Retrieval pools use $N \leq 1000$ unless noted; the null is empirical. Generation (Electronics-100K) uses gpt-5-mini with a gpt-5.2 judge. We report retrieval Recall/Precision/F1 and mean k^* ; for end-to-end runs, Token-F1, Title-F1, and Fuzzy-Title-F1; and an LLM judge over context precision/recall, answer faithfulness, and correctness.

3 Earlier Exploration: Why a Controlled Setting

Before the Amazon study we integrated HC into three established pipelines and ran it end to end. The purpose of summarising those results here is not to claim them as the contribution, but to explain why they pushed us toward a controlled environment.

HC inside Adaptive-RAG (BM25). On a keyword-retrieval (BM25 / Elasticsearch) backend with a shared gpt-4o-mini reader, HC *won* on **CrossEntityQA**, the dataset whose relevant-document count varies most widely (2–20 per query): passage-level F1 0.620 versus single-step *one*r 0.609 and multi-step IRCot 0.540. On **MuSiQue** HC was *competitive* (F1 0.301, within noise of IRCot’s 0.315). On **SQuAD** HC *failed* — and the failure was instructive: the stored null distribution overlapped 99.2% of the evaluated questions, so the p -values were uniform *by construction*, the

Method	Selection concept	Adaptive k ?	Backend used	Reference
Top- k	fixed budget	no	dense / BM25	—
Adaptive-RAG (router)	classifier routes nor/oner/IRCoT	yes (discrete)	BM25	(Jeong et al. 2024)
Adaptive-K (gap)	largest similarity gap	yes	dense / BM25	(Labs 2025)
FlashRAG (host)	modular pipeline; HC as post-processor	via cutoff	dense	(Jin et al. 2025)
HC (ours)	argmax of HC statistic over p -values	yes	dense / BM25	(Donoho and Jin 2004)
Berk–Jones	KL-divergence variant of HC	yes	dense	(Berk and Jones 1979)
Bonferroni	FWER, flat α/n threshold	yes (degenerate)	dense	(Dunn 1961)
Benjamini–Hochberg	step-up FDR control	yes	dense	(Benjamini and Hochberg 1995)
CAR (cluster)	agglomerative clustering on (rank,dist.)	yes	dense	(Xu et al. 2025)

Table 1: Selection rules evaluated in this project. HC and Berk–Jones are the only rules optimal on the Donoho–Jin detection boundary; the classical corrections (Bonferroni, BH) use flat thresholds; Adaptive-K and CAR ignore the null entirely.

HC statistic collapsed to ≈ 0 , and HC degenerated to a constant k . This was our first direct evidence that the null, not the HC math, is the fragile component.

HC inside Adaptive-K / FlashRAG. Across long-context aggregation (HoloBench) and multi-hop QA (HotpotQA), the qualitative pattern was consistent: HC matched or beat the gap-based Adaptive-K heuristic on answer quality while retrieving far fewer documents and tokens, but trailed the multi-step IRCoT router on dependency-heavy questions where evidence must be gathered across reasoning hops. On BEIR FiQA, where relevant passages are *scattered* rather than clustered at the top, HC underperformed; a cross-encoder rerank followed by a fixed threshold recovered +7.5% recall and +81% precision over a Top-5 baseline, foreshadowing the hybrid direction of Section 7.

The limitation these share. Every one of these settings entangled HC’s behaviour with a confound we could not control: a keyword-only backend (no embedding semantics), a tiny clustered corpus that invalidated the cross-query null (CrossEntityQA), a fixed candidate pool, or a surrounding router / reasoning loop. None let us vary *one* property of the data and watch HC respond, and none had fully known ground truth at the document level. To test *when HC works and when it fails*, we needed a realistic but controllable corpus: large, genuinely sparse, with a ground-truth relevance tier for every document and a per-query true K . That is what the Amazon datasets provide.

4 The Amazon Controlled Study

4.1 Proof of Concept: Category-Key

Category-Key is engineered to satisfy HC’s assumptions. The 726-document corpus (506 relevant city passages, 220 distractors) has relevant passages that genuinely cluster at high cosine similarity, a per-query empirical null that is near-perfectly calibrated (pooled KS = 0.003, 44/44 queries pass), and a true K that varies across small/medium/large buckets so no fixed k can serve all queries. Here HC behaves exactly as theory predicts (Table 2): F1 0.893 at mean $k^* = 9.7$, beating every fixed top- k in the sweep, with the HC cutoff correlating 0.835 with the true K — direct evi-

Method	Recall	Precision	F1
Top-10 (best fixed- k)	0.800	0.766	0.783
Top-20	0.939	0.511	0.662
HC (mean $k=9.7$)	0.852	0.938	0.893

Table 2: Category-Key: HC beats every fixed- k on F1 by adapting k per query (true- K correlation 0.835). This validates the HC machinery in the regime its assumptions hold.

dence that the argmax tracks the relevance boundary rather than collapsing to a constant.

4.2 Turning the Difficulty Knob: Compound Queries

We then introduced **compound** queries that combine a category filter with a numerical price constraint (e.g. “Electronics priced above \$150”). This is a deliberate, measurable violation of HC’s premise: text embeddings match category semantics but **cannot reason about numerical inequalities**, so hard negatives (right category, wrong price) sit right next to relevant documents in cosine space. To scale, we also adopted a **single global null** — one calibrated distribution shared by all queries — replacing Category-Key’s per-query nulls. Under that null HC breaks on Amazon-Compound: **55 of 135 queries return a negative HC statistic and select $k=0$** . Because relevance and true K are known, we can ask precisely *which* 55, and *why*.

4.3 Diagnosing the Failure: Three Mechanisms

The 135 queries partition exactly into three mechanisms.

(1) Embedding limitation (24 “irreducible”). The price signal is simply absent from the cosine: relevant documents are scattered through the ranks (similarity gap ≈ 0), below the Donoho–Jin detection boundary. Figure 3 shows a failing query (left) where relevant green points are strewn across ranks 50–195, versus a working query (right) where they concentrate at the top. No statistical test can recover the left case — the pipeline failed before HC ran.

(2) Calibration mismatch (31 “fixable”). Here the embedding *does* separate, but the per-query non-relevant z -score spread ($\sigma_q \approx 1.74$) is much narrower than the global null ($\sigma_{\text{null}} \approx 2.06$). A narrower true spread compared against a

Group	n	Top-50	Global null	+ rescale
positive_hc	77	0.233	0.253	0.251
fixable	31	0.164	0.000	0.157
irreducible	24	0.014	0.000	0.012

Table 3: Per-group macro-F1 on Amazon-Compound. `rescale_trim` recovers 30 of 31 fixable queries (0.000 $\rightarrow$ 0.157), preserves the 77 already-working queries (within 0.002), and correctly leaves the 24 embedding-limited queries untouched — eliminating the dominant failure mode without labels and without touching the null.

wider null inflates every p -value, so relevant documents look less anomalous than they are and the negative-HC gate trips. The std-ratio mismatch is the *dominant* predictor of HC failure ($r = 0.83$ with the HC statistic). *This is the “ p -values are too high” problem raised in our presentation feedback*, localized to a concrete, measurable cause.

(3) Signal dilution (general). With a fixed scan window γ , the HC win condition $p_{(i)} < i/N$ tightens linearly as the pool N grows, so a larger pool is not a free fix: it trades calibration gains for dilution losses (Figure 7 shows weak-signal queries never clearing the i/N bar).

The key realization: the 31 fixable queries are broken by *miscalibration*, not missing signal — and the miscalibration is a direct consequence of the global-null constraint we adopted for scale. The fix must therefore recalibrate per query *without* abandoning the single global null.

4.4 The Breakthrough: `rescale_trim`

`rescale_trim` is a label-free, per-query transform applied to the z -scores *before* the global-null lookup. For each query we estimate a robust noise scale $\hat{\sigma}_q$ as the standard deviation of the pool z -scores *after dropping the top γ ranks* (where relevant documents concentrate and would otherwise contaminate the estimate), then rescale:

$$z'_i = z_i \cdot \frac{\sigma_{\text{null}}}{\hat{\sigma}_q}, \quad \hat{\sigma}_q = \text{std}(z_{(\gamma+1:N)}).$$

This maps every query onto the same scale as the global null, so the empirical-CDF lookup becomes meaningful and the $r = 0.83$ std-ratio mismatch is cancelled directly. The transform uses only observable pool statistics (no labels) and is applied uniformly to every query — the null *itself* stays a single global distribution, so the architectural constraint is preserved. Table 3 shows the effect by group.

4.5 p -value Calibration: a Direct Response to Feedback

HC presumes that irrelevant-document p -values are uniform under the null. The calibration-mismatch failure mode is precisely a violation of that assumption: when $\sigma_q < \sigma_{\text{null}}$, the p -values of a query’s bulk are stochastically *too large* (skewed away from uniform), which both suppresses the HC statistic and explains why a naive global null made HC behave *conservatively* — selecting too few documents and depressing recall, the opposite of HC’s expected liberal

Method	R	P	F1	Mean k
Top-20 (fixed)	0.179	0.192	0.185	20.0
Top-50 (fixed)	0.287	0.139	0.187	50.0
HC + rescale	0.209	0.209	0.209	34.0
Berk–Jones + rescale	0.238	0.176	0.202	34.8
Adaptive- k (2025)	0.104	0.204	0.138	9.0
CAR (2025)	0.598	0.060	0.110	416.5
Benjamini–Hochberg	0.064	0.087	0.074	1.6
Bonferroni	0.014	0.049	0.022	0.3

Table 4: Algorithm comparison on Amazon-Compound (135 queries, pool $N = 1000$). HC posts the best F1 and highest precision; Berk–Jones follows closely.

profile. `rescale_trim` restores uniformity-under-null per query: after rescaling, each query’s bulk z -scores match the null’s spread, the empirical CDF returns near-uniform p -values for irrelevant documents, and only genuinely relevant documents retain small p -values. Figure 4 shows the clean-regime reference (Category-Key, where per-query nulls are already uniform); the same uniformity is what `rescale_trim` re-establishes on the compound corpus.

4.6 Final Results and Algorithm Comparison

On Amazon-Compound, HC + `rescale_trim` is the *only* method whose F1 strictly beats every fixed top- k , at an adaptive mean $k = 34.0$, and it leads a panel of eight selection rules on the identical pipeline (Table 4, Figure 5). The classical corrections collapse (flat thresholds cannot find an adaptive cutoff); CAR reaches HC’s F1 only when its pool is hand-tuned to $N = 100$, whereas HC needs no pool-size hyperparameter.

4.7 Generalization to Scale and to Generation

On the larger single-department Electronics-100K (100K docs, 44.5% uncapped hard negatives) HC again attains the best retrieval F1 (0.192 vs. 0.185 top-20, 0.165 top-50), with +34% recall over top-20 at adaptive $k = 34$. The advantage carries through the full RAG chain (Figure 6): against top-20, HC lifts **Token-F1** +14.3%, **Title-F1** +9.2%, and **Fuzzy-Title-F1** +15.4%, reaching 94–97% of top-50’s generation quality with 32% fewer documents — the adaptive- k benefit is not a small-dataset artifact.

5 Contributions

1. **A controlled-environment methodology for HC.** A family of Amazon retrieval datasets with fully known relevance tiers, per-query true K , and a single difficulty knob, enabling failure to be *attributed* rather than guessed.
2. **A failure-mode decomposition.** On compound queries we separate HC failures into embedding limitation (24), calibration mismatch (31), and signal dilution, and identify the std-ratio mismatch ($r = 0.83$) as the dominant, fixable cause.

Failure Mode Q2: Embedding Cannot Encode Price-Threshold Constraints
Relevant docs (green) sit ABOVE the price line; FAISS rank should sort them to the top — but it doesn't on irreducible queries

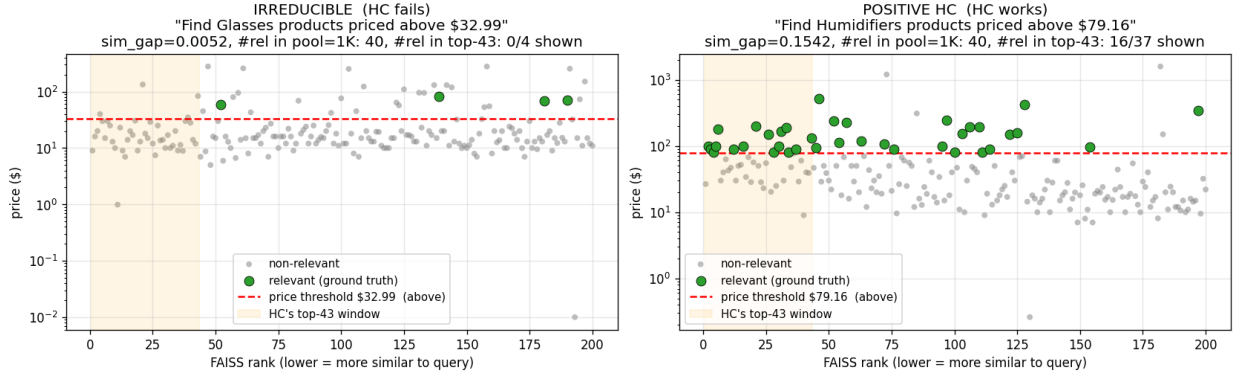


Figure 3: Embeddings cannot encode price. Each panel plots the top FAISS candidates for one query (x: rank, y: price; green = relevant, red dashed = price threshold, orange band = HC’s window). *Left (irreducible)*: relevant documents are scattered through the ranks — the discriminative signal (price) is not in the cosine. *Right (works)*: relevant documents concentrate at the top and HC fires.

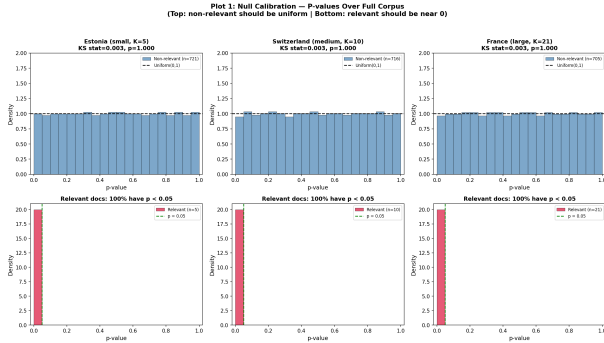


Figure 4: Null calibration in the clean regime (Category-Key): the per-query null is near-uniform (pooled KS = 0.003), so HC’s p -values are trustworthy. On compound queries this uniformity is lost under a single global null and is what `rescale_trim` restores.

3. **rescale.trim**. A label-free per-query recalibration applied before a single global null that restores p -value uniformity, recovers 30/31 fixable queries, and makes HC the best of eight selection rules — directly addressing the “ p -values too high” feedback.
4. **Best-in-class adaptive cutoff across regimes**. HC is the top method on Category-Key, Amazon-Compound, and Electronics-100K, generalizing from 726 to 100K documents and through end-to-end RAG.
5. **Pipeline integrations and a sparse benchmark**. Drop-in HC cutoffs in Adaptive-K, Adaptive-RAG, and FlashRAG, and the CrossEntityQA sparse multi-entity benchmark used to stress-test adaptive- k selection.

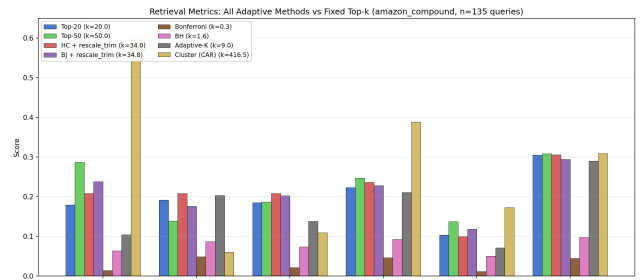


Figure 5: Retrieval metrics across all eight selection rules on Amazon-Compound. HC and Berk–Jones occupy the precision/recall frontier; FWER/FDR rules under-retrieve, CAR over-retrieves.

6 Discussion

When HC wins. Sparse-relevance corpora with well-separated embeddings and clustered relevant passages, where the number of relevant documents is both large and variable (Category-Key, Amazon-Compound after recalibration, CrossEntityQA). Here adaptivity converts directly into F1 and large token savings.

When HC loses. (i) *Embedding-limited* queries, where the discriminative signal is absent from the cosine (compound price constraints) — a pipeline failure upstream of HC. (ii) *Dependency-heavy multi-hop* questions, where iterative reformulation beats a single cutoff. (iii) *Miscalibrated nulls*, which silently turn HC conservative — the cause we localized and fixed.

Response to presentation feedback. Table 5 maps each comment to where it is addressed. The central technical

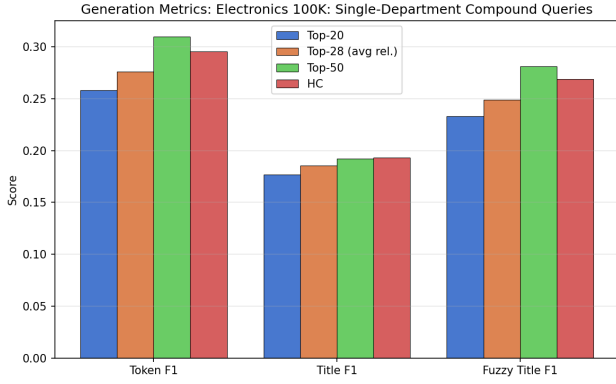


Figure 6: Electronics-100K end-to-end generation F1 (gpt-5-mini reader). HC’s retrieval advantage transfers to generation quality at a fraction of top-50’s document budget.

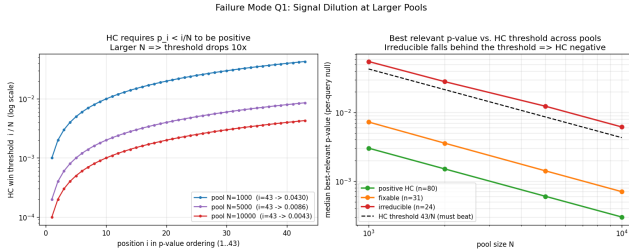


Figure 7: Signal dilution. As the pool N grows, the HC threshold i/N at a fixed window drops $\sim 10\times$; positive-group queries clear it but weak-signal (irreducible) queries never do — the empirical Donoho–Jin boundary.

comment — that our precision/recall profile was inverted because p -values were too high under a flawed null — is not merely acknowledged but *diagnosed* (Section 4.5) and *repaired* (Section 4.4).

7 Conclusions and Future Work

Treating “how many documents to retrieve” as a sparse-signal detection problem, we showed that Higher Criticism is a competitive query-adaptive cutoff — but only as good as its null. Our controlled Amazon datasets let us validate HC where its assumptions hold, decompose its failure on hard compound queries into three mechanisms, and design `rescale.trim`, a label-free per-query recalibration that eliminates the dominant (calibration) failure mode while respecting a single global null. The result is the best retrieval F1 of any method tested, generalizing from 726 to 100K documents and through to end-to-end RAG quality. The residual “irreducible” failures are a property of the embeddings, not of HC. Three directions follow:

- **Hybrid retrieval.** Combine embeddings for category semantics with structured metadata filtering for the numeric constraint, removing the embedding-limited failure mode at its source.

Feedback	Addressed in
Frame retrieval as sparse signal detection	§1.1, Fig. 1
Explain the p -value of a (query, doc) pair, visually	§1.2, §2.1
Illustrate at least one full RAG pipeline	Fig. 2
Table of pipelines tried (concept, adaptivity, ref.)	Table 1
Separate methodology from results	§2 vs. §4
Use graphics for results	Figs. 3–7
P/R inverted; p -values too high; verify uniform-under-null	§4.5, §4.4

Table 5: Presentation feedback and where each point is incorporated.

- **Rerank + HC.** Apply HC after a cross-encoder rerank to address scattered relevance; the FiQA threshold result is a positive existence proof.
- **Iterative HC.** Embed HC inside a multi-step retrieval loop to combine multi-hop coverage with adaptive-cost control.

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Appendix A: Project Roadmap

Week	Milestone / Outcome
1–2	Baseline RAG over CRAG (Meta AI 2024); full-document retrieval; k -sweep establishes fixed- k reference points.
3	RecursiveChunker (384 / 50 overlap); BGE bge-base-en-v1.5 selected as default embedder.
4	First HC integration into our pipeline; per-query null calibration.
5–6	HC into Adaptive-RAG (BM25): CrossEntityQA win, MuSiQue competitive; SQuAD failure traced to a same-query null (99.2% overlap) — first evidence the null is the fragile part.
7	HC into Adaptive-K and FlashRAG; HoloBench / HotpotQA / FiQA exploration; efficiency benchmarks.
8	FiQA scattered-relevance diagnosis; cross-encoder rerank + threshold workaround (+7.5% R, +81% P).
9	Decision to move to a controlled environment; design of the Category-Key dataset (known tiers, variable K).
10	Category-Key results: HC F1 0.893 beats every fixed- k ; k^*-K correlation 0.835.
11	Amazon-Compound (80K, 135 compound queries) built; global-null constraint adopted; 55/135 queries fail.
12	Failure-mode research: embedding-limited (24), calibration-mismatch (31), dilution; std-ratio $r=0.83$ identified.
13	rescale_trim designed and validated; 30/31 fixable queries recovered; HC leads eight selection rules.
14	Electronics-100K generalization (retrieval + end-to-end); final presentation; this report.